**AGA KHAN UNIVERSITY EXAMINATION BOARD**

**HIGHER SECONDARY SCHOOL CERTIFICATE**

**CLASS XI**

**ANNUAL EXAMINATIONS (THEORY) 2024**

**Business Mathematics Paper II**

**Time: 1 hour 5 minutes Marks: 20**

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**INSTRUCTIONS**

**Please read the following instructions carefully**.

1. Check your name and school information. Sign if it is accurate.

| **I agree that this is my name and school.**<br>**Candidate's Signature** |
|---|
| |

**RUBRIC**

2. There are SIX questions. Answer ALL the questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.

6. You may use a scientific calculator if you wish.

## List of Formulae

**Note**:

- The symbols have their usual meanings.

**Interest and Annuities**

$I = \dfrac{PTR}{100}$ $\quad A = P \times \left(1 + \dfrac{r}{n}\right)^{nt}$ $\quad R = \left(1 + \dfrac{i}{n}\right)^{n} - 1$

$j = \left(1 + \dfrac{r}{m}\right)^{m} - 1$ $\quad$ Accumulating factor $= \left(\dfrac{(1+i)^n - 1}{i}\right)$ $\quad$ Annuity $= R \times \left(\dfrac{(1+i)^n - 1}{i}\right)$

**Linear Equations, Functions and their Graphs**

The general form of linear equation is $ax + by + c = 0$

The intercepts form of the linear equation is $\dfrac{x}{a} + \dfrac{y}{b} = 1$ $a \neq 0$ and $b \neq 0$

**Quadratic Equations, Functions and their Graphs**

Quadratic Formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

**Derivative of Algebraic Functions**

$\dfrac{d}{dx}[c] = 0$ $\quad \dfrac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \times f'(x)$

$\dfrac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$ $\quad \dfrac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$

$\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$

**Sequence and Series**

$a_n = a_1 + (n-1)d$ $\quad S_n = \dfrac{a_1(1 - r^n)}{1 - r}$, if $|r| < 1$ $\quad S_n = \dfrac{a_1(r^n - 1)}{r - 1}$, if $|r| > 1$

$S_n = \dfrac{n}{2}(2a_1 + (n-1)d)$ $\quad a_n = a_1 r^{n-1}$

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Q.1. (Total 4 Marks)

In a factory, the time required in hours ($h$) is directly proportional to the number of items ($n$) produced and the time required in hours ($h$) is inversely proportional to the number of workers ($m$).

i. Using the given information, form an equation for the given variation. (2 Marks)

ii. It takes 12 hours for 30 men to complete a demand of 60 items of a certain product. Find the value of the constant. (2 Marks)

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**(ATTEMPT EITHER PART a OR PART b OF Q.2.)**

Q.2. (Total 4 Marks)

a. Find the effective rate of interest corresponding to nominal interest rate of 10% per annum compounded

i. annually. (2 Marks)

ii. quarterly. (2 Marks)

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**(ATTEMPT EITHER PART a OR PART b OF Q.2.)**

b. Saad deposits Rs 400,000 in his savings account at a bank. The account earns 4% interest per year, compounded semi-annually.

What will be the accumulated amount in his account after 8 years?

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Q.3. (Total 3 Marks)

Arham bought *x* dozen bananas for Rs 600. If he could pay Rs 10 less per dozen, he would have been able to get a dozen more.

By converting the above statements into mathematical expressions, show that the quadratic equation is formed as $x^2 + x - 60 = 0$.

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**(ATTEMPT EITHER PART a OR PART b OF Q.4.)**

Q.4. (Total 3 Marks)

a. Solve the given simultaneous equations.

$$3x^2 + y^2 = 4$$
$$x - y + 2 = 0$$

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**(ATTEMPT EITHER PART a OR PART b OF Q.4.)**

b. Maximise the objective function $Z = 2x + 3y$ subjected to the constraints $x + y \leq 3$, $2x - y \leq 0$, $x \geq 0$ and $y \geq 0$.

For the given linear programing problem,

i. shade the feasible region on the given graph.

ii. write the corner points.

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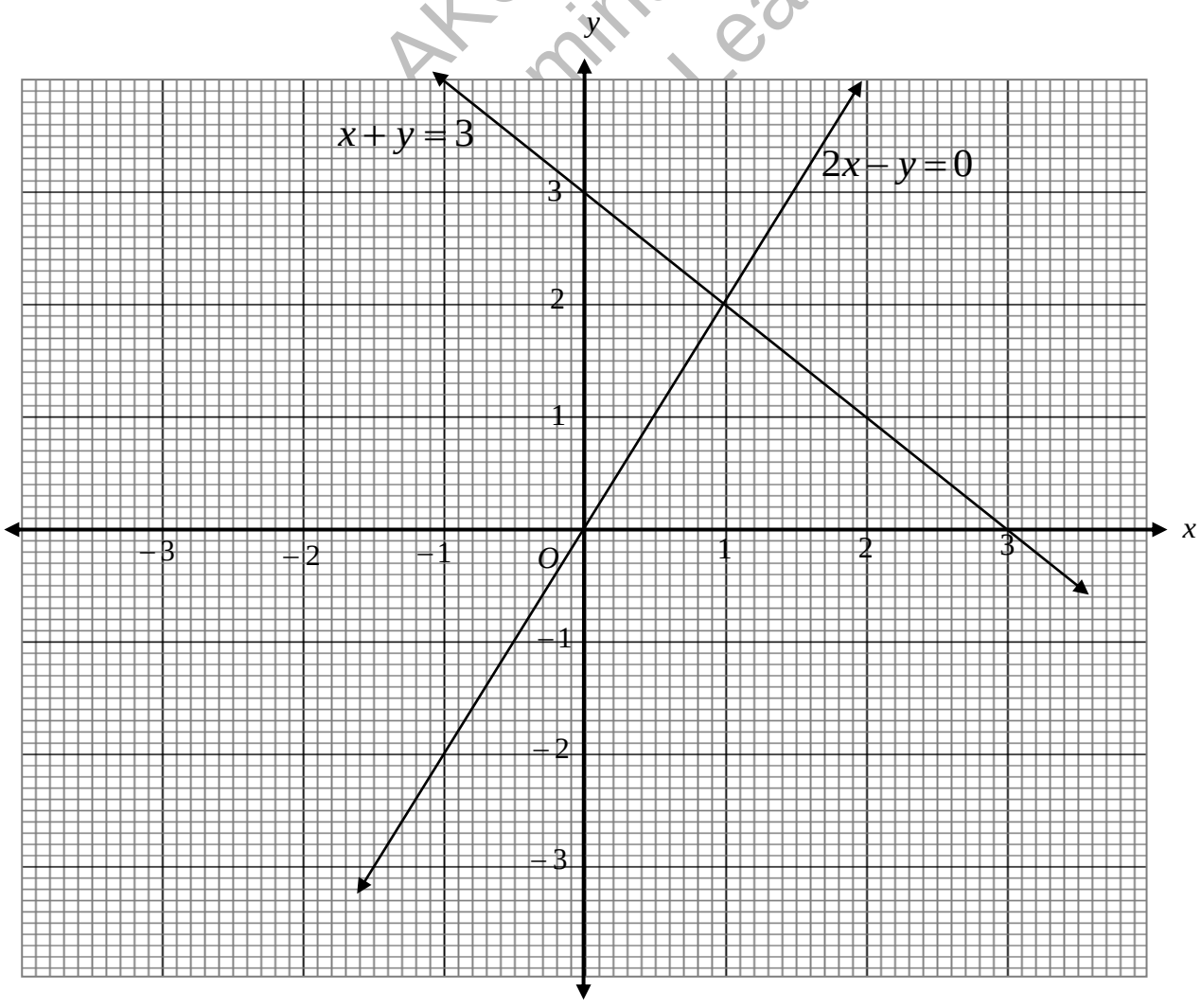
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iii. find the maximum value of the function.

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Q.5. (Total 3 Marks)

Find the value of $x$ by solving the given simultaneous linear equation using Cramer's rule.

$$x + y = 4$$
$$2x + y = 7$$

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**(ATTEMPT EITHER PART a OR PART b OF Q.6.)**

Q.6. (Total 3 Marks)

a. Find the sum of the given arithmetic series.

$$1 + 3 + 5 + 7 + 9 + \ldots + 103$$

b. The sum of the finite geometric series is 1.2222 and $r$ is equal to 0.2. Find the first term of the series.

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END OF PAPER

# Please use this page for rough work

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